

b)

```
function sum(a, b) {  
  return a + b;  
}
```

```
sum(1, '2');
```

c)

```
<div onclick="console.log('div')">  
  <p onclick="console.log('p')">  
    Click here!  
  </p>  
</div>
```

D)

```
const numbers = [1, 2, 3];  
numbers[10] = 11;  
console.log(numbers);
```

E)

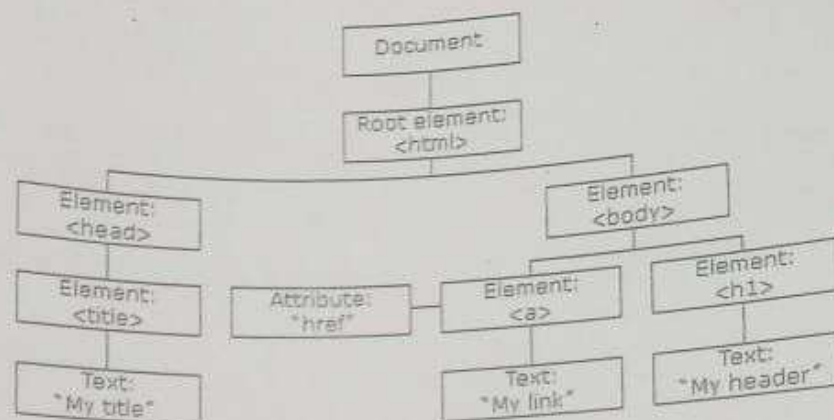
```
console.log(3 + 4 + '5');
```

F)

```
console.log(018 == '018')  
console.log(017 == '017')
```

CLO # 2: Apply web programming principles to create responsive and functional, interactive web applications.

Q3: write the code in JavaScript using the concept of DOM to create the leaf nodes of the tree below?
[5 marks]

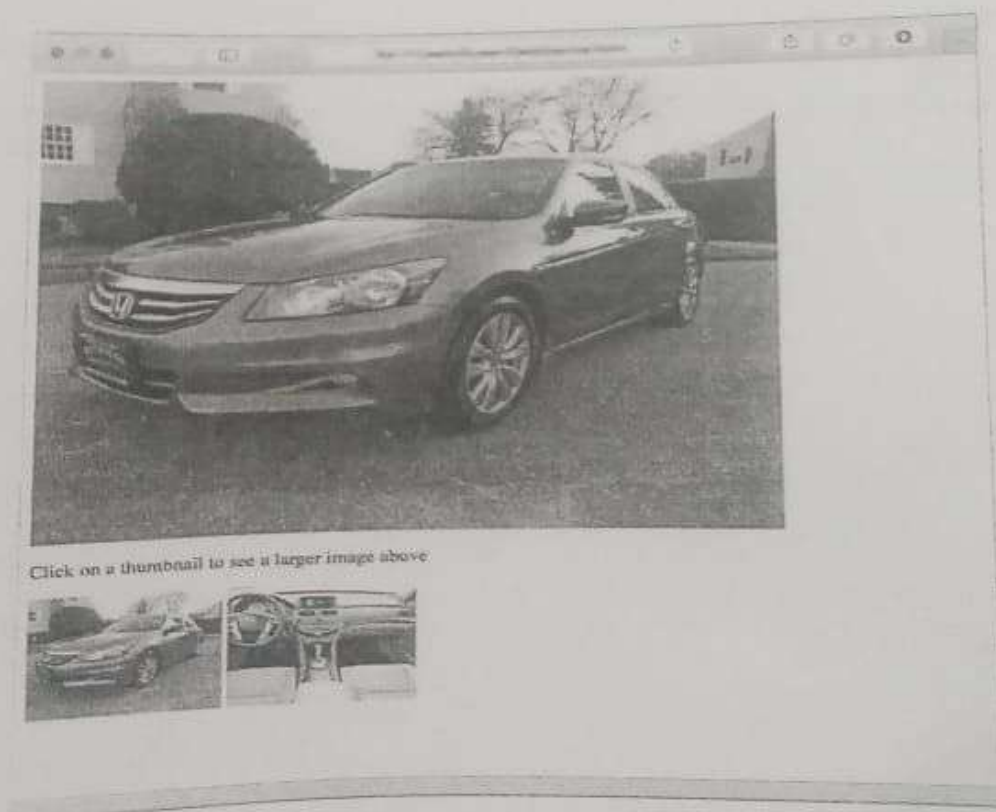


CLO 3: Analyze user requirements and web development problems to design solutions

Q4:

[15 marks]

Ali wants to sell his car online. He has taken a couple of pictures of his car and he wants to provide a page where people can take a look at his car. Here is what his web page looks like:



And this is the relevant part of his code so far:

```
</p>  
<p>  
    
  
  Click on the thumbnail to see a larger image above.  
  <br>  
  <img src='exterior_small.jpg' class='thumb' data-big='exterior_large.jpg'>  
  <img src='interior_small.jpg' class='thumb' data-big='interior_large.jpg'>  
</p>
```

Now, replicate Ali's project in **React**, so that if someone clicks on the small thumbnail of the exterior shot, the large exterior shot gets displayed in the big image at the top of the image (i.e. the image with the id 'car').

Similarly, if the small thumbnail of the interior shot is clicked, the larger interior shot gets displayed in the big image at the top of the page. Assume that the code (component) is defined in a file `car.js`.
Code for `car.js`:

CLO 3: Analyze user requirements and web development problems to design solutions

Q5:

[15 marks]

Write the code in **Node JS** for a web service `books.js` that outputs JSON data about books available at a book store in a particular category. This service should take a GET parameter named `category` and search a data file for all books in that category. Your Node JS code will read a data file named `books.txt`.

Each line in that file matches the following format, with each token of information separated by a single vertical bar:

name|author|category|date|price

All tokens of data are guaranteed not to contain any vertical bars in them. You can assume all data in the file is valid although it may contain extra blank lines and these shouldn't crash your program or alter your output.

The JSON that is output should contain a number labeled `"count"`. This should represent a count of all books at the book store. It should also contain a list called `"books"` that contains a list for each book in the passed in category. The list for each book should contain the year labeled as `"year"`, the price labeled as `"price"`, the title labeled as `"title"`, and the author labeled as `"author"`. If the category parameter is not passed return a 400 status and stop execution immediately. If no books match the category you should send back the same data as usual, just leave the course array empty. An example input file:

21 Burgers for the 21st Century|Stuart Reges|cooking|2010|75.00
Harry Potter and the Sorcerer's Stone|J.K. Rowling|children|1998|19.99
Breakfast for Dinner|Amanda Camp|cooking|2009|22.00

The expected output for the provided input file when running Books.js with the query parameter category=cooking is:

```
{ "count":3,  
  "books":[ {"year" : "2010",  
              "price" : 75.00,  
              "title" : "21 Burgers for the 21st Century",  
              "author" : "Stuart Reges"  
            },  
            {"year" : "2009",  
              "price" : 22.00,  
              "title" : "Breakfast for Dinner",  
              "author" : "Amanda Camp"  
            }  
          ]  
}
```

CLO 3: Analyze user requirements and web development problems to design solutions

Q6:

[15 marks]

Write JavaScript code in a file called simple.js for manipulating a list. The page UI allows the user to type some text into a text box. The user can click an "add" button to make the text appear in the list (put it inside the game div).

When the mouse enters a list item the background color of that item should turn cyan and the font size of that list item should increase by 5px. When it leaves that item the background color should turn back to white and the font size back to what it originally was. The original font size is included in a CSS file attached to the HTML. Your code should work for any starting font size value. You can view the current size with your code but not by opening the file and looking at it.

When the user clicks on an item it should be deleted.

You may not use any Javascript libraries such as Prototype and JQuery. You must write ALL Javascript to make the page work, none is provided. You can define any additional CSS you like as long as you label it in a clear way. You only need to write the appearance/CSS changes mentioned above. You can assume the page has a linked style sheet that provides the other styles visible in the screenshots.

CSL	add
-----	-----

The page when it first loads

CSL	
154	
GREAT	
add	add

The page after a couple adds with the mouse hovering

CSL	
GREAT	add

The page after "154" and "Great" have been clicked and removed

Your Javascript will be manipulating the following HTML code:

```
<!DOCTYPE html>
<html>
  <head>
    <script type="text/javascript" src="simple.js"></script>
    <link rel="stylesheet" type="text/css" href="simple.css" />
  </head>
  <body>
    <div id="game"></div>
    <input id="input" type="text" />
    <button id="add">add</button>
  </body>
</html>
```

CLO # 1: Demonstrate understanding of fundamental web technologies (HTML, CSS, JS).

Q1:

[35 marks]

- A) Why did Internet gained so much popularity? Give answer in points. (5)
- B) What are the processing Islands in the enterprise and how we can judge a software application based upon them? Give answer in points. (5)
- C) What is the difference between Web and Internet? Give answer in points. (5)
- D) Explain what this URL directs to:
<http://www.funwebdev.com/index.php?page=17#article> (5)
- E) what is the difference between hoisting, scope chaining, lexical scope and closure in JavaScript? (5)
- F) Why there was a need of JQuery, when we already had JavaScript to solve the problems? Give answer in points. Give example of sample code. (5)
- G) what is the difference between bubbling and capturing events in JavaScript? Explain with the help of diagram? (5)

CLO # 1: Demonstrate understanding of fundamental web technologies (HTML, CSS, JS).

Q2: what is the output of following code and give proper reason?

[15 marks]

a)

```
for (var i = 0; i < 3; i++) {  
  setTimeout(() => console.log(i), 1);  
}
```

```
for (let i = 0; i < 3; i++) {  
  setTimeout(() => console.log(i), 1);  
}
```